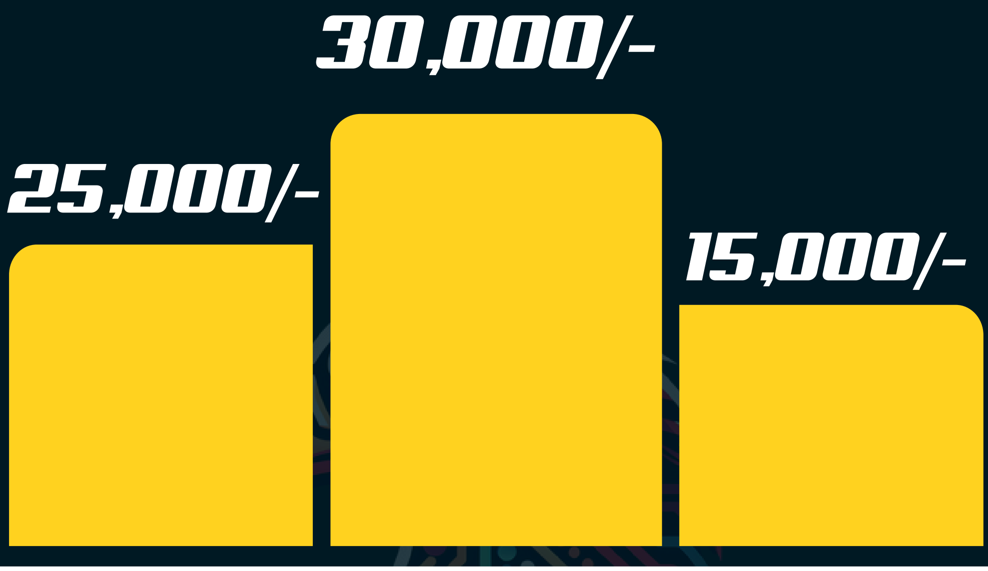
ROBO RASHTRA '26

RESQLYMPICS 3.0

RESQLYMPICS

Resqlympics is a robotics challenge where a manually operated bot must navigate through a closed, obstacle filled arena using only a live camera feed. The objective is to rescue hostages of three distinct types, each associated with specific time constraint and point values, and securely place them in a carrier attached to the bot. The event emphasises precision, control, time management and strategic decision-making in determining the order of rescues.



PRIZE POOL

70,000/-

BOT SPECIFICATIONS

SIZE RESTRICTIONS:

- Length = 300 mm
- Width = 300 mm
- Height = 250 mm
- Ground clearance = 30 mm to 40 mm
- Max bot weight = 2.5 kg

A margin error of ± 20 mm is acceptable.

MANUAL CONTROL REQUIREMENTS:

- All teams require manual control for operating the robot during the competition.

CONTROL MECHANISM:

- Teams must employ a wireless control mechanism.
- No size restrictions for the remote-control mechanism.
- Only one team member is allowed to control the robot.

BOT SPECIFICATIONS

CONTROL MECHANISM DESIGN:

Control mechanism must contain screen (e.g. mobile, laptop, etc.) on which the live feed from bot is taken. The robot's control mechanism should be designed for operation by a single person. Controller and the streaming device both can be separate.

ADHERENCE TO SPECIFICATION:

Failure to comply with any of the specified regulations will result in immediate disqualification from the competition.

POWER SUPPLY:

- The robot must rely solely on an onboard power supply, such as rechargeable batteries or power cells, and cannot rely on external power sources during the competition.
- Voltage output = Maximum 12.5V
- Safety protocols must be followed during battery recharging.

GAME PLAY

Starting Position:

- Robots will enter the arena from the start point.
- Before the official start, participants can take a brief observation period (e.g., 2 minutes) to strategise and plan their robot's path for hostage rescue.
- Once the game officially starts, participants cannot make further strategic decisions.

Game Duration: The total gameplay time is 10 minutes

Rescuing Hostages:

- Robots must identify and reach hostages placed in different areas of the arena. Size of each hostage is 30x30x30 (mm) and max 150 g
- A rescued hostage must be securely stored by the robot in its carrier only.
- The specific hostage must be rescued within its specified time constraint.
- The operator can choose which hostages to rescue first based on the time conditions provided for each type of hostage.

GAME PLAY

Obstacle Handling:

- Obstacles include sand, speed breakers, marble pit, bridge, curves, blocks, etc.
- Robot must navigate through obstacles without causing damage to the arena, or hostages and reach the end point.
- Penalties may be incurred (e.g. deduction of points) for any violations, including improper handling of hostages.
- Any aggressive or unsafe behaviour will result in penalties (for eg. unnecessarily annoying and arguing with event officials.)

Emergency Stop:

- If a robot gets stuck or encounters an obstacle preventing further movement, it may be moved back to previous checkpoint by the event officials for which penalty would be added.

- The robot can resume its operation from that checkpoint, but no extra time will be allotted for the setback.
- Time will continue to run uninterrupted during this process and will not be paused during emergency stops.

Rounds:

The ResQlympics event will be held in a single day, comprising of two rounds.

- Evaluation Round: Teams will present their bots to the judges. Judges will evaluate the teams based on the specifications and features of the bot.
- Gameplay: The actual rescue mission will be performed in this round.

The team that rescues the hostages with maximum points will emerge as the winner.

GAME PLAY

Example: Consider the arena contains three types of hostages: R, G and Y, each with its own time constraint (lifetime) and points. The robot may rescue any hostage only while its time window is active. Once the time limit of any hostage type expires, that hostage is considered **dead** and the robot **must not pick it up**. The team must then focus on rescuing the remaining hostage types within their active time windows and complete the competiton within the overall game time.

Winners:

- The participant or team with the highest total points at the end of the event will be declared as winner.
- In case of a tie in total scores, the winner will be selected based on the overall completion time of the bot.

TEAM SPECIFICATIONS

A team may consist of 2 - 5 members.

GAME RULES

- Robots must adhere to size limitations and technical specifications as provided in the competition guidelines.
- All robots and control mechanisms will undergo compliance check before the challenge begins.
- Rescue operations must follow designated time limits
- Hostages must be rescued within their time constraint.
- The points of hostages picked after their time constraint will not be considered.
- Once a hostage is secured, the robot must securely/safely store it in its carrier.
- Safety protocols must be followed at all times to prevent harm to participants, spectators or the equipment.
- Once the competition begins, no manual interventions will be allowed(or else penalties will be added).

- Judges' decisions regarding scores, compliance and rules are final and binding.
- Failure to comply with the rules or any unsportsmanlike conduct will lead to immediate disqualification.

NOTE: The rules outlined in this rulebook are subject to change. Any modifications or updates to the rules will be communicated to all participating members prior to the start of the event